CSIT545 Final Example Questions – Chapters 4 (from pipelining) + Chapter 5 **December 2025**

MULTIPLE CHOICE

1) For the following code, with forwarding how many cycles will be stalled to avoid data hazard?
   or $a, $b, $c
   or $d, $a, $b

   A) 0
   B) 1
   C) 2
   D) 3

2) For the following code, what are the hazards?
   add $1, $1, $2
   add $3, $3, $4
   add $5, $5, $6
   add $7, $7, $8
   A) Data hazard only
   B) Control hazard only
   C) No data hazard, no control hazard
   D) Both data and control hazards

3) For the following code, without forwarding how many cycles will be stalled to avoid data hazard?
   lw $a, 0($b)
   or $c, $a, $b

   A) 1
   B) 2
   C) 3
   D) 0

4) Same code as in Question 5). With forwarding how many cycles will be stalled to avoid data hazard?

   A) 1
   B) 2
   C) 3
   D) 0

5) For the 5 pipeline stages:
   EX: Execute
   MEM: Memory Access
   ID: Instruction Decode
   IF: Instruction Fetch
   WB: Write Back

   What is the correct order?

   A) IF, ID, MEM, WB, EX
   B) IF, ID, EX, MEM, WB
   C) IF, MEM, ID, WB, EX
   D) EX, MEM, ID, IF, WB

6) lw $t1, 0($t0)
lw $t2, 4($t0)
add $t3, $t1, $t2
sw $t3, 12($t0)
lw $t4, 8($t0)
add $t5, $t1, $t4
sw $t5, 16($t0)

For 5-stage pipeline, the above code has load-use data hazard ($t2 and $t4), and has to stall. With stalls, how many cycles to excute this code?

A) 12
B) 11
C) 14
D) 13

7) For 5-stage pipeline, if each stage takes 100 picosecond (ps), then 10 instructions needs ____ ps.

A) 1100
B) 1400
C) 1200
D) 1000

8) Suppose a for-loop program uses 2 registers and loops 12 times. Using Loop-unrolling technique, to reduce the program to loop 4 times, you would need ____ registers.

A) 8
B) 12
C) 6
D) 2

9) For a MIPS dual-issue pipeline processor, the very long instruction word (VLIW) has ____ bits.

A) 32
B) 128
C) 64
D) None of above

10) What is an example of temporal locality?

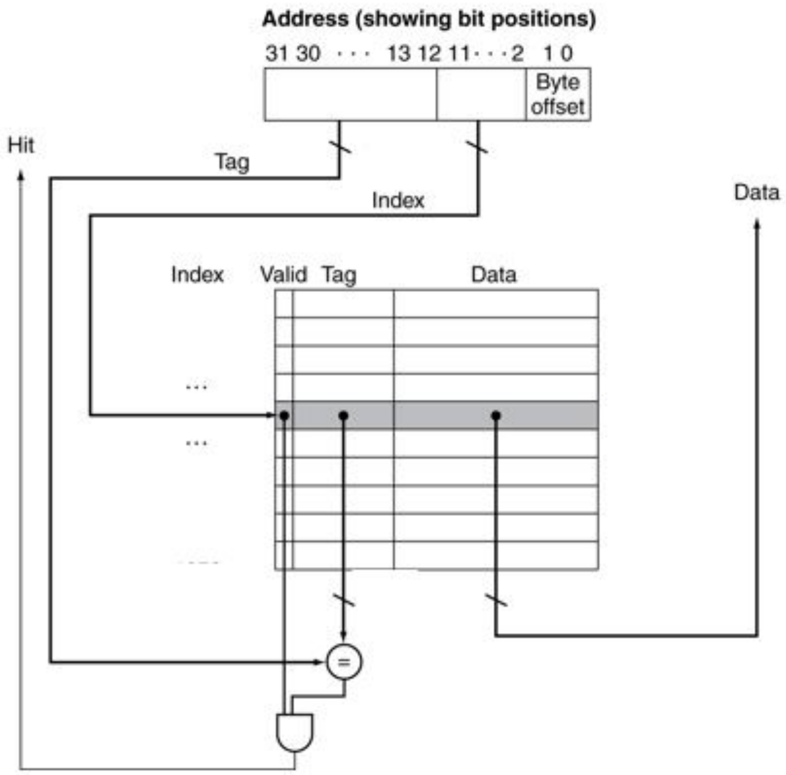
A) Array of integers
B) Sequential instruction access
C) Instructions in a loop
D) None of above

11) What is the fastest to slowest order of the following memory module?

A) SRAM, Solid State drive, DRAM, Magnetic drive
B) DRAM, SRAM, Solid State drive, Magnetic drive
C) SRAM, DRAM, Solid State drive, Magnetic drive
D) SRAM, DRAM, Magnetic drive, Solid State drive

12) For a Direct-mapped cache of 8 one-word, the memory word at Mem[01011] goes to Cache[?], with Tag = ?

A) Cache[11], Tag = 010
B) Cache[01], Tag = 011
C) Cache[010], Tag = 11
D) Cache[011], Tag = 01



13)

The above cache has ____ entries.

A) 2048
B) 1024
C) 512
D) 256

14) Each entry has____bits.

A) 52
B) 53
C) 54
D) 55

15) Memory address 10110010 10100001 00000000 11111111 is in cache entry____

A) 0000111111
B) 0011111111
C) 11111111
D) 00000000

16)

Bit position		1	2	3	4	5	6	7	8	9	10	11	12
Encoded data bits		p1	p2	d1	p4	d2	d3	d4	p8	d5	d6	d7	d8
Parity bit coverage	p1	X		X		X		X		X		X	
	p2		X	X			X	X			X	X	
	p4				X	X	X	X					X
	p8								X	X	X	X	X

An SEC Parity-coverage scheme is given above. Determine the 12-bit Hamming Error Correction Code given a 1-byte data value of 1111 1111.

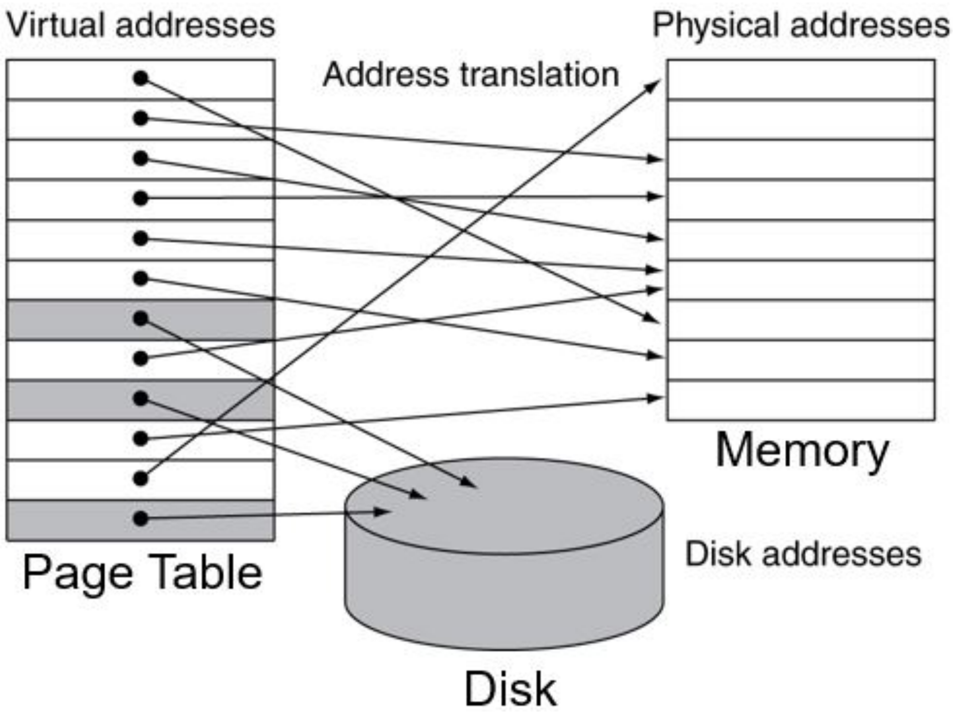
A) 111011101111
B) 111111111111
C) 111111101111
D) 111011111111

Two-way set associative

Set	Tag	Data	Tag	Data
0				
1				
2				
3				

17)
For the 8-word, 2-way set associative cache above, if the two consective addresses are 5 (Mem[0101]) and 13 (Mem[1101]), then

A) They are both in Set #3
B) They are both in Set #1
C) They are in Set #2 and Set #3
D) They are in Set #1 and Set #3



18)

In the virtual memory system shown above, which statement is correct?

A) Number of entries in Page Table > Number of pages in Disk
B) Number of entries in Page Table = Number of pages in Disk
C) Number of entries in Page Table = Number of pages in Memory
D) Number of entries in Page Table < Number of pages in Disk

19) Which of the following is an example of Spatial locality?

A) Recursive calls
B) Instructions in a loop
C) Accessing array data
D) None above

Answer Key
Testname: CSIT545FINAL_FA2025-SAMPLEQUESTIONS

1) A
2) C
3) B
4) A
5) B
6) D
7) B
8) C
9) C
10) C
11) C
12) D
13) B
14) B
15) A
16) A
17) B
18) B
19) C

CSIT545 Final Exam - Sample Open-ended Questions

1. The following MIPS code is executed in the pipelined MIPS:

```
lw  $t1, 0($t0)       # load b
lw  $t2, 4($t0)       # load c
add $t3, $t1, $t2     # get b + c
sw  $t3, 8($t0)       # store a
```

which accomplishes C/Java statements:

```
a = b + c;
```

(a) Supposing no forwarding, the code has data hazards. Identify where the hazards are, and insert the minimally needed `nop` instructions ("bubbles") to eliminate all data hazards.

(b) Supposing forwarding is implemented, the code still has but fewer data hazards. Identify where the hazards are, and insert the minimally needed `nop` instructions to eliminate all data hazards.

Answer:

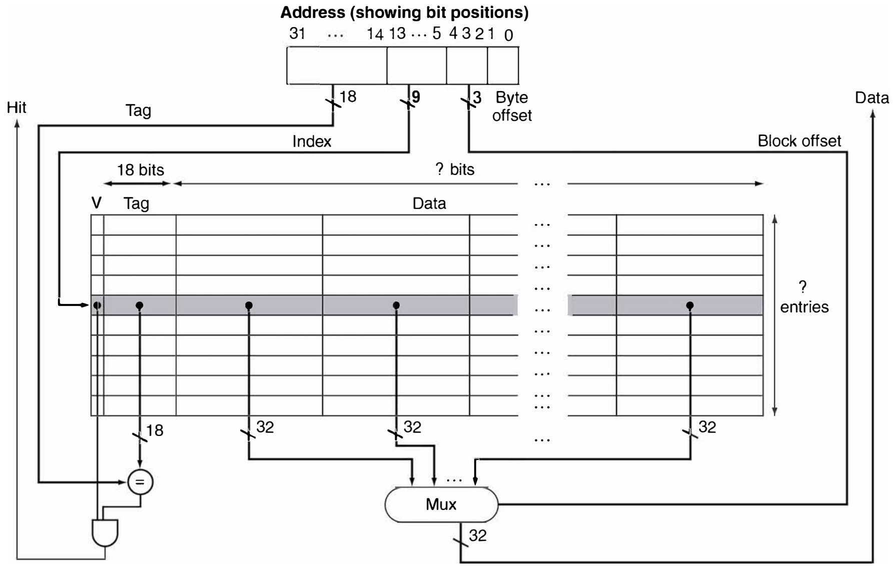
(a)

```
lw  $t1, 0($t0)       # load b
lw  $t2, 4($t0)       # load c
nop
nop
add $t3, $t1, $t2     # get b + c
nop
nop
sw  $t3, 8($t0)       # store a
```

(b)

```
lw  $t1, 0($t0)       # load b
lw  $t2, 4($t0)       # load c
nop
add $t3, $t1, $t2     # get b + c
sw  $t3, 8($t0)       # store a
```

2. Given the multi-word-block cache below:



(a) How many entries (blocks) does the cache have?

(b) How many words in each block?

(c) How many total number of bits in each block?

(d) For word address 1200, what is its *block address*?

Answer:

(a) $2^9 = 512$

(b) $2^3 = 8$

(c) $32 \times 8 + 18 + 1 = 275$

(d) $1200 \div 8 = 150$